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Review Article

Diffusion bonding, brazing and resistance welding of zirconium alloys: a review



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ABSTRACT

The joining of zirconium alloys is an important issue in the field of nuclear energy since zirconium alloys are often used as nuclear fuel cladding tube. The article reviews the research progress on the diffusion bonding, brazing and resistance welding of zirconium alloys. The main focus of this work is on the process parameters, interfacial microstructure and mechanical properties of joint. For diffusion bonding, the process parameters including temperature and interlayer have significant influence on the properties of joint. The range of temperature and the selection of interlayer was reviewed and the influence of them on interfacial microstructure and properties of joints was also discussed. For brazing, the influence of brazing filler, temperature and holding time on the properties of joint was summarized. For resistance welding, the main process parameters include welding current and contact resistance. Numerical simulation method was often used to describe the influence of process parameters on microstructure during welding end plug with cladding tube. Some suggestions for the further research direction on the welding and joining of zirconium alloys were also proposed.

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1. Introduction

As high-performance materials, zirconium and zirconium alloys have been widely used to fabricate many crucial parts in fields of nuclear and chemical industries based on their good corrosion resistance and small neutron absorption cross section [1–6]. The zirconium alloys are used as structural materials for fuel rods and spacer grids combined into fuel assemblies for nuclear reactors [7]. Nuclear grade zirconium as a national strategic metal plays an important role in the development of nuclear industry [8–10]. Most alloys are state-of-the-art versions of the Zr–Sn (Zircaloy-2 and Zircaloy-4) and Zr–Nb (E110, E125, M5, Zr-2.5%Nb) systems, or are multicomponent ones (E635, ZIRLO, NDA, MDA, etc.) [11–13]. Up-to-date data on the thermo-physical properties of zirconium and its commercial alloys are most fully summarized in. However, more and more zirconium alloys have been developed recently in order to meet the requirements for high performance of zirconium alloys in the industry engineering. The focus of designing and developing new zirconium alloys not only on changing the composition but also exploring advanced fabrication methods [14].

Welding as one of the most important processes for manufacturing the fuel assembly from the zirconium alloys has been attracted much attention in the field of nuclear energy. The excellent joining methods of zirconium alloys include bolts and welding. Fusion welding are main welding methods for zirconium alloys although some defects occur during welding process [14–17]. The joints obtained by fusion welding mostly have good properties. Gas tungsten arc welding (GTAW) is the most widely used welding process for zirconium alloys [18]. However, GTAW has an obvious drawback as low deposition rate and low productivity although some new types of GTAW process have been developed. Electron beam welding is also widely used for welding zirconium alloys in the nuclear industry and a lot of results of studies have been published to date [17,19–22]. Electron beam welding has demanding requirements for equipment and vacuum [23]. Laser welding is also well known for its ability to deliver deep penetration welding and another advantage is lack of mandatory use of vacuum chamber when compared with electron beam welding [16,18,24–29]. It should be noted a possible violation of stable heat generation upon welding due to absorption of laser beam energy by metal plasma clouds evaporated from the weld pool and a change in the reflection coefficient at the edge surfaces [23]. Researchers have done much work on the fusion welding of zirconium alloys and the summary and review on fusion welding of zirconium alloys have also done by researchers [14,23,30].

During manufacturing the nuclear components, it is required that many kinds of welding technology are adopted together on an assembly. As shown in Fig. 1, the design of welded adapters of process channels for a reactor includes GTAW, electron beam welding and diffusion bonding three main welding methods [31]. Diffusion bonding, brazing and resistance welding are becoming more and more significant in the field of welding zirconium alloys.

Diffusion bonding refers to the solid-state joining method with press at high temperature without visible deformation

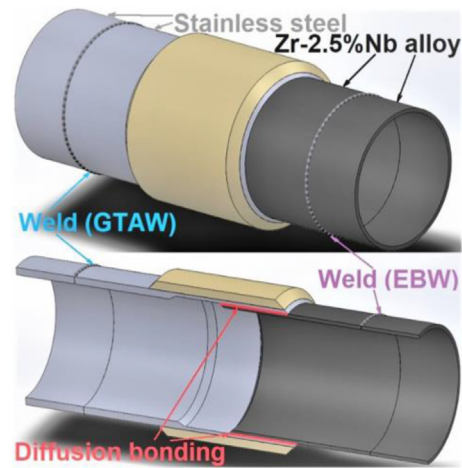


Fig. 1 – A typical design of welded adapters of process channels for reactor bolshoy moshchnosty kanalny (RBMK) reactors [31].

and relative movement. Diffusion bonding has been widely used in manufacturing reactor fuel parts, electronic components, corrosion-resistant parts [32,33]. Brazing refers to the joining method that the brazing filler with the melting point below that of the base material and the workpiece are heated to the melting temperature of the brazing filler. And the liquid filler is filled into the gap of the solid workpiece to wet and join base material. Brazing has been widely used in manufacturing cemented carbide tools, drilling bits, bicycle frames, heat exchangers, conduits and various containers, motors,

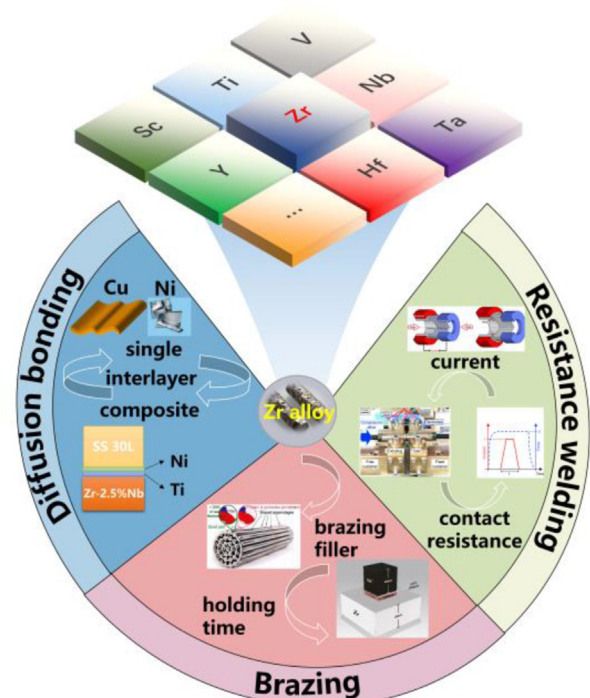


Fig. 2 – Overview of diffusion bonding, brazing, resistance welding of zirconium alloys.

instruments and rad [34]. Resistance welding is an effective welding method by using the impedance of metals themselves and the contact impedance between metals, causing local thermal melting with the help of current, and applying certain pressure at the same time. It is suitable for zirconium alloys to be joined by resistance welding because of its high conductivity, thermal conductivity and low thermal expansion [35,36]. Zirconium alloy is also used as the cladding of nuclear reactor because of its excellent properties [37]. One of the fabrication techniques of the cladding of nuclear reactor is resistance welding [38].

The main focus of present work is to summarize the progress on diffusion bonding, brazing and resistance welding of zirconium alloys, as shown in Fig. 2. The advantages, application, process parameters, properties of joints of above three joining methods will be reviewed in the following part, which will provide useful information for selecting joining method or optimizing the process as well as help develop new joining method for zirconium alloys.

2. Diffusion bonding of zirconium alloys

The high melting point of zirconium alloy make it critical to select the appropriate processing parameters such as temperature, interlayers, pressure and holding time during diffusion bonding zirconium alloys. The reasonable selection of these process parameters can improve the quality and properties of the joint. In order to analyze the internal reasons, it is necessary to study the interfacial microstructure and bonding mechanism of the joint.

2.1. Influence of process parameters on properties of joint

The selection of process parameters often has a great influence on the properties of the joint. Choosing the appropriate temperature and the interlayer can improve the properties of the joints. The interlayer is becoming more and more useful in diffusion bonding application. Designing the composition and type of interlayers could decrease the melting point and avoid the interfacial intermetallic compounds during diffusion bonding.

2.1.1. Influence of interlayer on properties of joint

During diffusion bonding zirconium alloys to other different alloys, it was easy to form some brittle intermetallic compounds near the bonding interface. In order to minimize these intermetallic compound layers, Ti, Ni, Cu, Al, Nb, Ag and other metals are usually used as the interlayer. In addition, two or three metals can be used together as the composite interlayers, and even the thickness of the interlayers can be controlled. The type, composition and thickness of interlayer has significant effect on the properties of diffusion bonded joints. As listed in Table 1, the shear strength of diffusion bonded joints with different interlayers is summarized.

Single foil is usually considered as ideal interlayer for diffusion bonding zirconium alloy. Ni interlayer was often used to diffusion bond zirconium alloys. When diffusion bonding zirconium alloy to 304 stainless steel, the thickness of the reaction layer is only 55 μm in the joint without interlayer, while the thickness of the reaction layer increases to 135 μm in the joint with Ni interlayer [49], as shown in Fig. 3. In addition, cracks are formed in the joint while no cracks are detected in the joint with Ni interlayer, which can be explained by the fact that Ni as the interlayer increases the thickness of the reaction layer and enhances the relaxation effect of residual stress. Similarly, Batra et al. [50] used Ni as an interlayer to diffusion bond Cu–Cr–Zr alloy to stainless steel. No pores or cracks are found in the joint. The properties of the joint are improved and the good metallurgy is obtained.

Cu is an important commercial metal. Generally speaking, as an interlayer, Cu is used at a temperature that affects its properties during diffusion. Due to the alloying of the surface, the mechanical properties of the joint will be influenced significantly by the change of interfacial composition caused by diffusion. Zeng et al. [51] proved that the diffusion bonding of Zr705 alloy was finally realized by adding Cu as the interlayer at 900°C–960 °C. Cu is a good thermal conductor with high thermal conductivity and diffusivity, which allows higher heat flux. Therefore, Meng et al. [43] also used Cu as interlayer to realize diffusion bonding of pure zirconium and low carbon steel. Yang et al. [42] attempted to diffusion bond Zircaloy-4 to TiAlC with Cu as interlayer and the temperature was suggested to be controlled at 900 °C. It was found that ZrC was formed near the interface instead of brittle ZrAl, which enhanced the strength of the joint.

Table 1 – Shear strength of joints with different interlayers.

Base metal	Interlayer	Thickness of interlayer	Shear strength (MPa)	References
Zircaloy-4/Zircaloy-4	Ni foil	5 μm	358	[39]
Zr-4 alloy/Zr-4 alloy	Ti foil	100 μm	294	[40]
Zirconium/Q345 steel	Al foil	500 μm	30	[41]
Zircaloy-4/TiAlC	Cu foil	10 μm	221	[42]
Zirconium/Low carbon steel	Cu foil	90 μm	88	[43]
Zircaloy-4/304 L stainless steel	Ti/Ag	50 μm	165.7	[44]
ZrC–SiC/304 stainless steel	Ti/Ni	Ni (200 μm)	97	[45]
Zircaloy-4L/stainless steel 304	Ti/Ni	Ti (10 μm)	209	[46]
		Ni (20 μm)		
		Ti (40 μm)		
Zr-4 alloy/316 stainless steel	Nb/Ni	–	380	[47]
Zr-2.5Nb zirconium alloy/304 L stainless steel	Nb/Ni	Nb (30 μm)	84.4	[48]
		Ni (45 μm)		

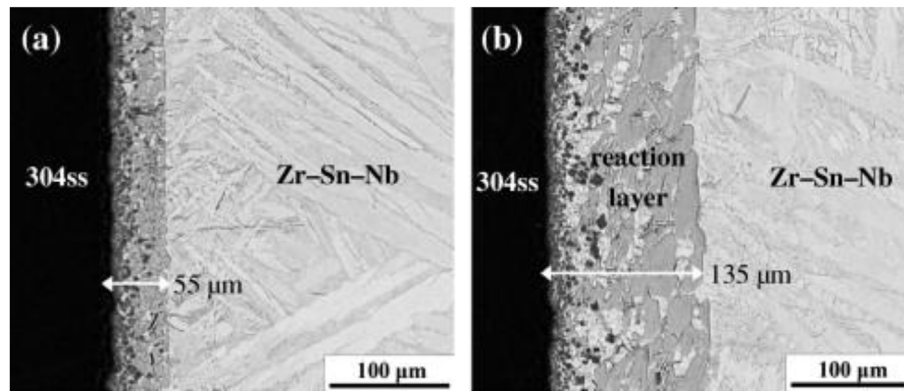


Fig. 3 – The cross-sectional microstructure of the joints bonded at 1000 °C for 30 min: (a) the joint without interlayer and (b) the joint with Ni interlayer [49].

Zhang et al. studied the influence of Al interlayer on the properties of diffusion bonded joint of zirconium alloy and Q345 steel [41]. The thicknesses of the reaction layer in both Al/Fe and Al/Zr interfaces are temperature-dependent, with a higher temperature yielding a larger thickness. The hardness of Al_5Fe_2 is higher than Al_3Zr and the base materials. The propagation of shear crack occurs within Al_5Fe_2 . Optimized shear strength can be obtained by controlling the thickness of reaction layer. During diffusion bonding zirconium-4 to 304 austenitic stainless steel [52], Fe interlayer was used to reduce the formation of intermetallic compounds and limit the diffusion of Ni into Zr alloy.

Akhter et al. used Ti as interlayer to realize diffusion bonding Zr-4 to 316 stainless steel [53]. Mazar Atabaki et al. also used Ti as interlayer for transient liquid phase (TLP) diffusion bonding zirconium alloy to stainless steel [54]. The optimum shear strength of 99 MPa with minimum deformation was found to be for specimens bonded at 850 °C for 15 min. Sun et al. used pure Ti as interlayer for diffusion bonding Zr-4 alloy [40]. After using the pure Ti foil interlayer, the bonding temperature was reduced by 100 °C to achieve the same shear strength of 294 MPa. Therefore, the pure Ti interlayer can help realize the low temperature diffusion bonding of Zr-4 alloy by interfacial phase transition inducement.

Many kinds of single interlayer have been tried to diffusion bond zirconium alloys and the influence of single interlayer on the properties of joints have been also revealed. However, the strength of joints is still expected to be improved. Composite interlayers provided a possible way to improve properties of joints by changing the interfacial microstructure. Srikanth used Ni and Ti as composite interlayer to study the diffusion bonded joints of zirconium alloy and 304 stainless steel [46]. When the temperature and holding time in the range of 800 °C–900 °C and 30–90 min, the shear strength of the joint changes between 70 MPa and 209 MPa as shown in Fig. 4. It is found that the optimum bonding parameters are 850 °C and 60 min under the pressure of 20 MPa. The interfacial intermetallic compounds are composed of NiZr_2 , NiZr , Ni_3Ti , NiTi and NiTiZr . It is found that the joint fails due to the formation and propagation of cracks passing through NiZr_2

intermetallic compounds layer during the shear test. It is also found that Ni and Ti as composite interlayer can reduce the residual stress of joints zirconium alloy and stainless steel.

Pan et al. [44] used Ti and Ag as composite interlayer to carried out diffusion bonding Zircaloy-4 and 304 L stainless steel at 800 °C under 12 MPa for 15–120 min. The shear strength of the joints increased firstly and then decreased with increasing bonding time, and the maximum shear strength of 165.7 MPa was reached for 90 min. Although there was TiAg IMCs formed in the joints, which did not lead to hardening of the joints. Wang et al. used Ni and Nb as composite interlayer for diffusion bonding of 304 stainless steel and Zr-2.5Nb at 850 °C–1050 °C under 4 MPa for 30 min [48]. And the joint assembly form is 304 stainless steel/Ni/Nb/Zr-2.5Nb. The maximum strength of 84.4 MPa was obtained at 1000 °C. Yang et al. [47] also used Ni and Nb as composite interlayer to diffusion bond Zr-4 alloy and 316 stainless steel.

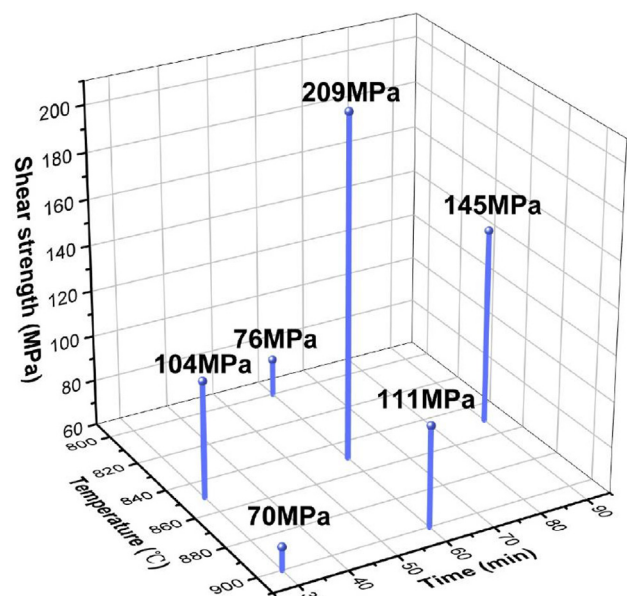


Fig. 4 – Shear strength of joint of zirconium alloy and 304 stainless steel [46].

The maximum shear strength of the joint without interlayer is 131 MPa at 900 °C for 30 min under 10 MPa. And the maximum shear strength is 380 MPa at 820 °C.

Many researchers have studied the influence of the type of interlayer on the properties of the joint. Lin et al. have explored the influence of the thickness of interlayer on the properties of the joint between TZM alloy and Nb–Zr alloy [55]. The temperature was set at 1000 °C–1100 °C with the holding time of 30 min to conduct diffusion bonding. It was found that at the lower temperature of 1000 °C, compared with without Ni interlayer, the Ni interlayer with a thickness of 30 µm could make the two base metals better bonded, and the shear strength of the joint was improved by 60%. In order to further improve the shear strength of the joint, the thickness of the Ni interlayer was increased to 45 µm, and it was found that the shear strength was obviously improved. When the temperature was raised to 1100 °C and the thickness of Ni interlayer was 45 µm, the shear strength of the bonded joint was 264 MPa. It was suggested that with the increase of the thickness of Ni interlayer, the thickness of the diffusion zone did not change obviously.

The thickness of the interlayer has a significant effect on the property of bonded joints and the property can be improved by properly increasing the thickness of interlayer. For diffusion bonding of zirconium alloy and other dissimilar alloys, it is possible to bond dissimilar alloys without the interlayer. In industry, it is important to reduce the processing cost, so it is also encouraged to carry out diffusion bonding without interlayer.

2.1.2. Influence of temperature on properties of joint

Temperature is an important factor affecting the interfacial diffusion behavior and bonding from both thermodynamic and kinetic aspects, so the reasonable selection of temperature is an important index to determine the properties of joints. Table 2 lists some results of studies on the temperature used for diffusion bonding zirconium alloys and other alloys.

Aboudi et al. studied the properties of joint of zirconium alloy and stainless steel with Cu as interlayer at the temperature range of 1173 K–1323 K [64]. The welding time was controlled for 45 min. It was found that at the temperature of 1173 K, half of Cu reacted with zirconium alloy, and the hardness of the reaction layer reached 547 HV. At the same time, cracks propagating perpendicular to the interface can be observed. When the temperature is 1223 K, the Cu

intermediate layer is completely dissolved, and the hardness of the reaction layer reaches 641 HV. Cracks can be observed at the interface of the reaction layer/diffusion affected zone, and the cracks are parallel to the interface. At the temperature of 1293 K, the thickness of the reaction layer is 150 µm, and the hardness of the reaction layer is 690 HV. At a higher temperature of 1323 K, the thickness of the reaction layer is 400 µm. The hardness value of the reaction layer reaches the maximum measured value of 792 HV.

It was found that the influence of temperature on the mechanical properties of the joint is not only reflected in the hardness of the joint, but also in the shear strength of the joint [65]. By controlling the temperature between 850 °C and 1000 °C, the solid diffusion of 304 L stainless steel and Zr-4 alloy was realized. The final test showed that the shear strength first increased and then decreased with the increase of temperature, and the maximum shear strength reached 194 MPa at 950 °C [65]. Effect of bonding temperature on mechanical properties of 304 L/Zircaloy-4 diffusion bonded joints with Ni/Ta composite interlayer as studied by Wang [66]. Ni₃Ta intermetallic compounds were formed at the Ni/Ta interface and the solid solution of Zr and Ta appeared at the Ta/Zr-4 interface. The shear strength of joints showed a trend from rising to dropping with the increase of the bonding temperature. The maximum shear strength reached 194 MPa and the highest hardness value across the interface of joints was 356 HV in Ni₃Ta layer.

The shear strength and the Ni₃Nb layer thickness of Zr-4/Nb/Ni/316SS joints was influenced by the bonding temperatures and holding times [47], as shown in Fig. 5. The shear strength of the joint first increased and then dropped significantly from 780 °C to 840 °C. The shear strength of the joint was maximum at 820 °C, reaching 380 MPa. However, as a result of the excessive thickness of the Ni₃Nb layer, the shear strength dropped to 342 MPa at 840 °C. As shown in Fig. 5(b), the shear strength increased first and then dropped down slowly as the holding duration prolonged. Diffusion bonding of Zr-2.5Nb zirconium alloy and 304 L stainless steel was carried out with Nb/Ni composite interlayer and with the increase of bonding temperature, the shear strength of the joint increased first and then decreased [48].

Zeng et al. [51] found that the tensile strength of joints increased with the increasing temperature, and the elongation first increased and then decreased with the increasing temperature. At 900 °C and 920 °C, the limited diffusion of Zr atoms and Cu atoms at lower temperatures led to the formation of intermetallic compounds in the joints. At 940 °C and 960 °C, Cu atoms completely diffused into the zirconium alloy, and no intermetallic compounds were observed. Coupling effect of bonding temperature and reduced interlayer thickness on the properties of the diffusion bonded joints of Zr alloys was also investigated [67]. It was also found that the thickness of the interlayer and the fracture mode of the joint is related to the temperature. The diffusion bonding of Zr705 alloy with Cu as the interlayer was carried out. It was found that the reduction of bonding temperature can reduce the thickness of the interlayer and improve the tensile properties of the joint. The fracture form of the bonded joints was changed from brittle fracture at 880 °C or 900 °C to ductile fracture at 920 °C or 940 °C.

Table 2 – Temperature for diffusion bonding zirconium alloys and other alloys.

Base material	Temperature	References
Zirconium alloy/stainless steel	1123 K–1223 K	[56]
Zirconium alloy/stainless steel	1023 K–1213 K	[57]
Zirconium alloy/Ni based alloy	1083 K–1173 K	[58]
Zirconium alloy/FeCrAl alloy	873 K–1073 K	[59]
ZrSnNb alloy/FeCrAl alloy	1123 K	[60]
Zr-4 alloy/Ti alloy	1373 K–1573 K	[61]
Zirconium alloy/316 stainless steel	1273 K, 1323 K	[62]
Zirconium alloy/aluminum alloy	698 K–898 K	[63]

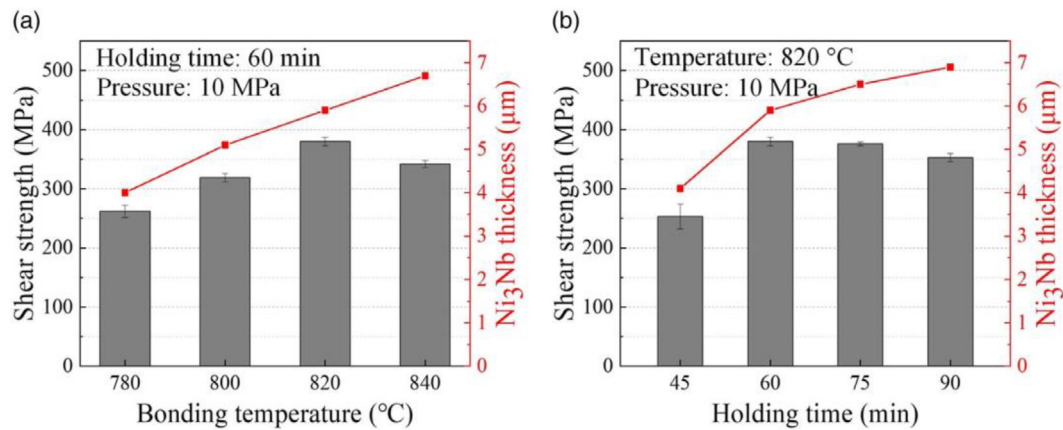


Fig. 5 – The variation of the shear strength and the Ni_3Nb thickness of Zr-4/Nb/Ni/316SS joints under 10 MPa with different (a) bonding temperatures and (b) holding times [47].

The effects of diffusion bonding temperature and holding time on the interfacial microstructure and mechanical properties of the diffusion bonded Zr-4 alloy joints were

investigated by Wang in detail [66]. Increasing temperature or prolonging the holding time can result in the coarseness of grains and the formation of brittle second phase. The interface

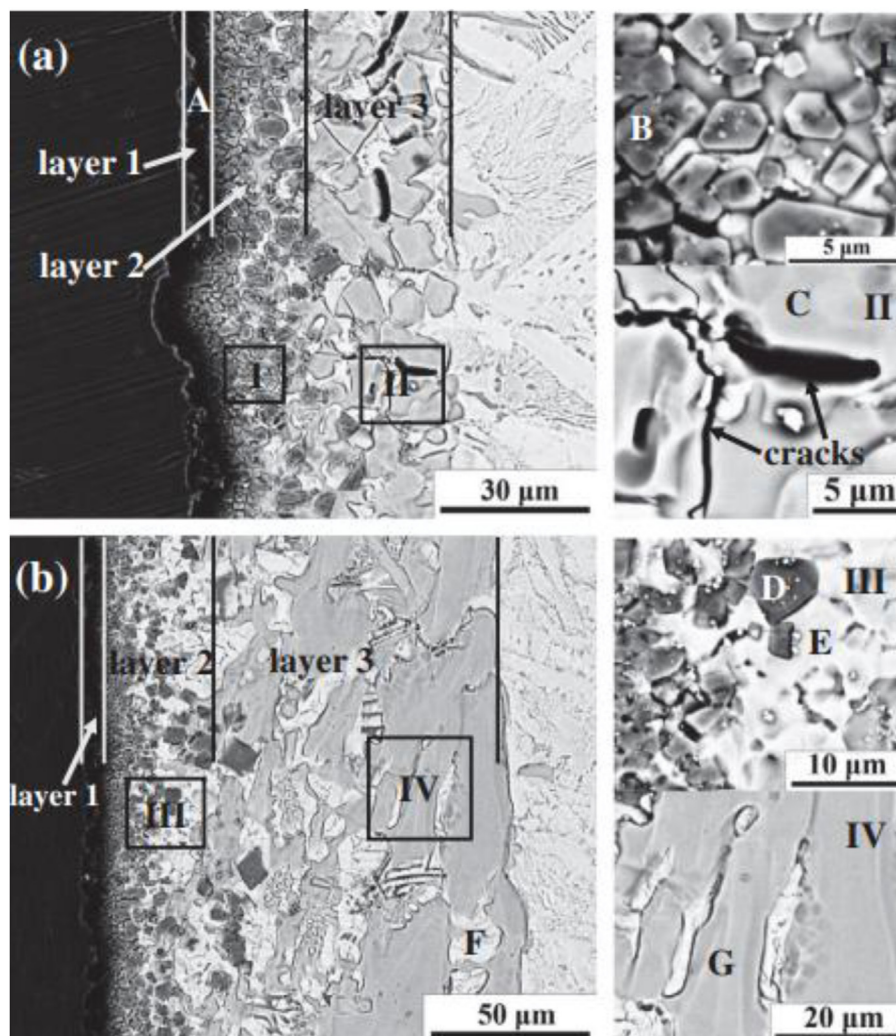


Fig. 6 – The microstructure of the reaction layers of the joints: (a) the direct bonded joint and (b) the joint with Ni as interlayer [49].

bonding ratio of the diffusion bonded Zr-4 joint gradually increased from 74% to 95% with the increasing of bonding temperature. The highest shear strength of 349 MPa was obtained at 800 °C for 30 min under 7 MPa.

The improvement and innovation in the processing technology of diffusion bonding can effectively decrease the bonding temperature. The diffusion bonding of Zr and Ti has been realized at an ultra-low temperature of 650 °C by nanocrystallising the substrate surfaces [68].

2.2. Interfacial microstructure and bonding mechanism

The properties of the diffusion bonded joint are usually affected by the interfacial microstructure directly, so it is necessary to study the interfacial microstructure and bonding mechanism. Diffusion bonding of 304 stainless steel and zirconium alloy with Ni as interlayer was studied [49]. The interfacial reaction layer is mainly composed of α -FeCr layer, $\text{Zr}(\text{Cr}, \text{Fe})_2 + \alpha$ -Zr layer and α -Zr + $\text{Zr}_2(\text{Ni}, \text{Fe})$ layer. Cracks are found in the reaction layer of joint without interlayer. Fig. 6 shows the comparison of the reaction layer between direct bonding and reaction bonding with Ni as the interlayer. The reaction layer in the joint with Ni interlayer is wider than that without interlayer, which is beneficial to the relaxation of residual stress.

Lebaili et al. [69] used diffusion bonding to join zirconium alloy and 304 stainless steel under the conditions of 820 °C and 950 °C, argon atmosphere and dynamic pressure. The interfacial metallurgical microstructure and its evolution were studied. The joint obtained at 950 °C has a reaction zone (RZ) formed at the stainless steel/zirconium alloy interface,

composed of three layers. The first layer (LI = α -(Fe, Cr)) on the stainless steel side and the third layer (LIII = $\text{Zr}_2(\text{Fe}, \text{Ni})$) on the zirconium alloy side are single-phased. The middle layer LII is biphasic (LII = $\text{Zr}(\text{Cr}, \text{Fe})_2 + \text{Zr}_2(\text{Fe}, \text{Ni})$).

Perona et al. studied diffusion bonding of zirconium alloy and stainless steel [70]. By analyzing the interface microstructure, they studied the intermetallic compounds at the interface of the joint and their distribution in the diffusion layer. The maximum strength of joint can be obtained when the diffusion layer thickness is 2 μm , because the intermetallic compounds formed can be ignored when the diffusion layer thickness is 2 μm . Taking Zn as the interlayer metal, Reboul et al. [71] studied the phase evolution of the interface between zirconium alloy and austenitic stainless steel.

Lebaili et al. [72] conducted sufficient research and experiments on the interface microstructure between 304 stainless steel and zirconium alloy at different temperatures. The temperature is controlled at 950 °C–1050 °C, and the isothermal holding time is controlled at 45 min. It was found that when heated to 950 °C, the microstructure of joint is mainly composed of three layers, namely α -FeCr, $\text{Zr}(\text{Cr}, \text{Fe})_2$, $\text{Zr}_2(\text{Ni}, \text{Fe})$. When the temperature is increased from 950 °C to 1050 °C, the thickness of the second layer decreases sharply and the thickness of the third layer increases significantly. During the isothermal process at 1050 °C, with the increase of holding time from 0 to 45 min, the thickness of the first layer increased slightly, and the thickness of the second layer and the third layer increased. Fig. 7 shows the microstructural evolution during the heating and isothermal process of stainless steel and zirconium alloy joint.

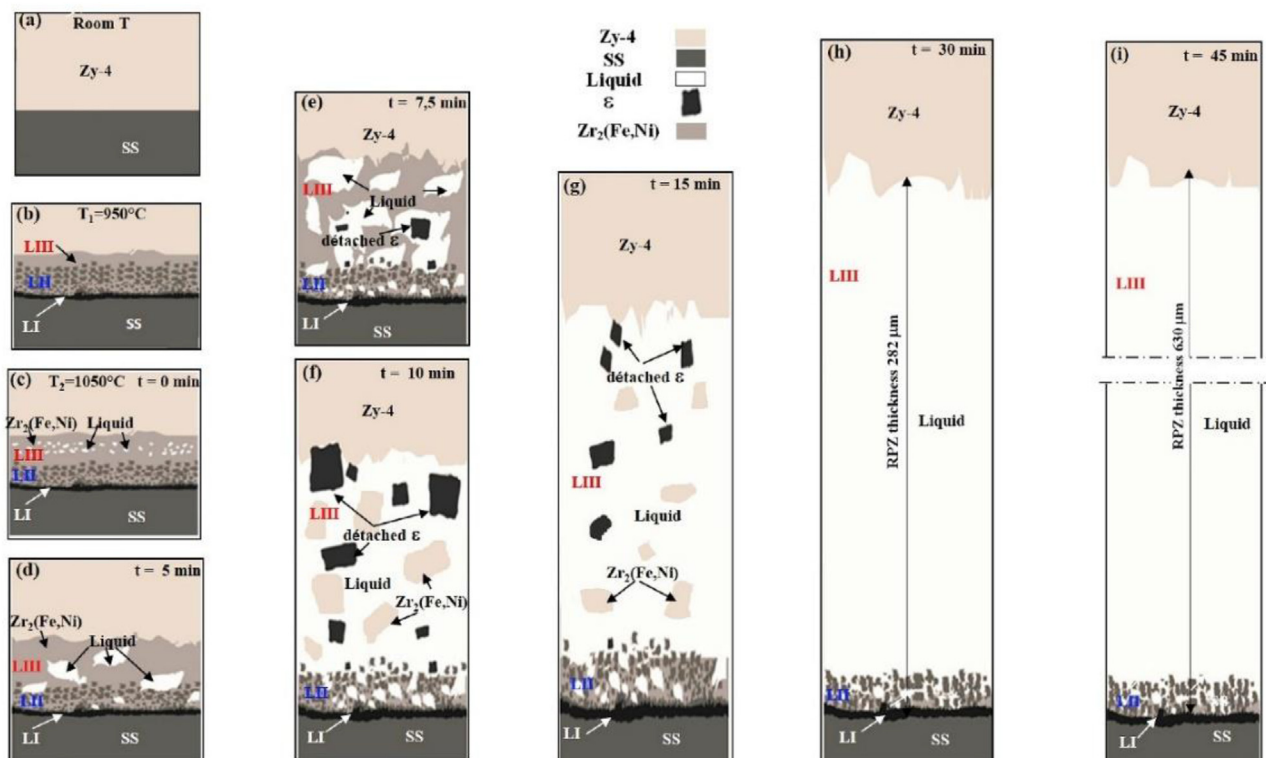


Fig. 7 – Stages schematic evolution of the stainless steel/zirconium diffusion couple during heating and isothermal holdings at 1050 °C [72].

Li et al. [73] studied the microstructure evolution and mechanical properties of Zircaloy-4/Cr joint using stainless steel as interlayer. It was found that of Cr and zirconium alloy the reaction zone of the direct diffusion bonded joint can be divided into two layers. The first layer is next to Cr, which is formed by diffusion of Cr and Zr. The second layer is α -Zr and Zr_2Cr obtained by eutectic reaction of Zr and Cr.

2.3. Interfacial bonding strength and fracture mechanism

The interfacial bonding strength of diffusion bonded joint is related with fracture mechanism. Therefore, both interfacial bonding strength and fracture mechanism will be discussed and reviewed in this part.

Zirconium alloy and 304 stainless steel was diffusion bonded and the force–displacement curve of the joint is shown in Fig. 8 [49]. Because there is no obvious plastic deformation, the fracture way is typical brittle fracture. Fracture of the direct joint occurs at the reaction layer. The compact intermetallic compounds, large residual stresses and cracks in the reaction layer may lead to the sudden rupture. The bonding strength of the joint bonded with Ni interlayer is higher than that without Ni interlayer. The bonding strength of the joint increases as a consequence of many α -Zr phases dispersedly existing in the reaction layer and wider reaction layer formed by using Ni interlayer.

The fracture morphology and mechanism of Zr705 diffusion bonded joint was investigated by Zeng et al. [51], as shown in Fig. 9. It was found that the fracture failure mode of welded joints at 900 °C and 920 °C was brittle due to the formation of intermetallic compounds. At 900 °C, intermetallic compounds were composed of Zr_7Cu_{10} , Zr_3Cu_8 and $Zr_{14}Cu_{51}$, while at 920 °C, intermetallic compounds were composed of Zr_2Cu , Zr_7Cu_{10} and $Zr_{14}Cu_{51}$. There are dimples and cleavage planes on the failure surface of the joint, so it can be inferred that the failure form of the fracture is ductile fracture. At 960 °C, it can be indicated that the fracture surface of the joint is ductile fracture because of the tearing ridge.

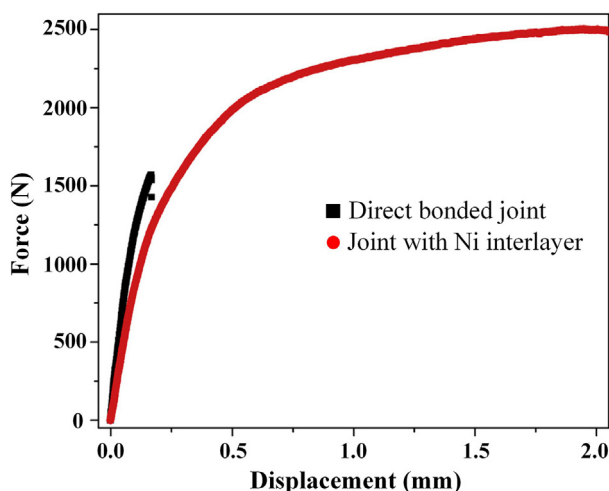


Fig. 8 – The force–displacement curves of the joints without and with Ni interlayer [49].

The effects of interlayer thickness on fracture cracks of joints were studied [55]. Without interlayer, the researchers found that diffusion zone with a thickness of 2.5 mm was formed between TZM and Nb–Zr, which provided a micro-hardness transition across the joint. The cracks during the shear test initiated from the (Mo, Nb) diffusion zone and then spread to TZM alloy. Even if the temperature was raised to 1100 °C, a complete joint without cracks could not be obtained.

Wang et al. studied the fracture morphology of Zr-4 alloy joint at different temperatures [66]. The fracture position of the joints presented different as at the interface and near base material. At 760 °C, as can be seen from Fig. 10, there is a large unbonded area near the crack, and the sample breaks along the interface, indicating that the joint failed to achieve effective bonding at the temperature of 760 °C. When the temperature rises to 800 °C, the unbonded area is not found in the joint and the fracture occurs near the base material, as shown in Fig. 10(c). When the temperature is 820 °C, some tearing ridges accompanied by river patterns can be observed in Fig. 10(d). The river pattern surrounded by tearing ridges is short and curved, which proves that the fracture is quasi-cleavage fracture mode.

Fig. 11 shows the comparison of fracture morphology under different holding time. Fig. 11(a) and (b) (30 min) show some river patterns, dimples and lacrimal ridges fractures, from which it can be inferred that it is quasi-cleavage fracture. When the holding time is prolonged to 90 min, as shown in Fig. 11(c) and (d), more second phase is observed than that with 30 min. By investigating the fracture surface of two samples, Zr(Cr, Fe) is considered as the source for crack initiation and propagation, so the shear strength of the joint decreases significantly with increasing holding time.

3. Brazing of zirconium alloys

Zirconium and its alloys are widely used in nuclear industry and chemical industry because of their high corrosivity and various excellent characteristics [74]. However, the dissimilar welding or joining between zirconium alloy and other materials is still a challenge in many cases. Brazing is an important joining method for zirconium alloys but the brazing of zirconium alloy has not attracted enough attention.

Some defects in brazed joint deteriorate the properties of joints. The unfilled space in the joint is called a gap. Generally speaking, the gap filled with gas and flux is more common in filler metal with slow flow [75]. In the process of brazing, the filler metal needs to interact with the base metal to form a metallurgical bond, but this may lead to the formation of intermetallic compounds that is not conducive to the mechanical properties of the joint.

In view of the brazing process of zirconium alloy, on the one hand, the defects of joint are encouraged to avoid. On the other hand, in order to make the brazed joint of zirconium alloy have good properties, it is necessary to conduct in-depth research on various factors affecting the brazing process. The reason why brazing can form a firm joint between base metals is because the brazing filler metal is added to join two different base metals. Under the heating condition, the brazing filler metal can be melted to join the base metals while ensuring

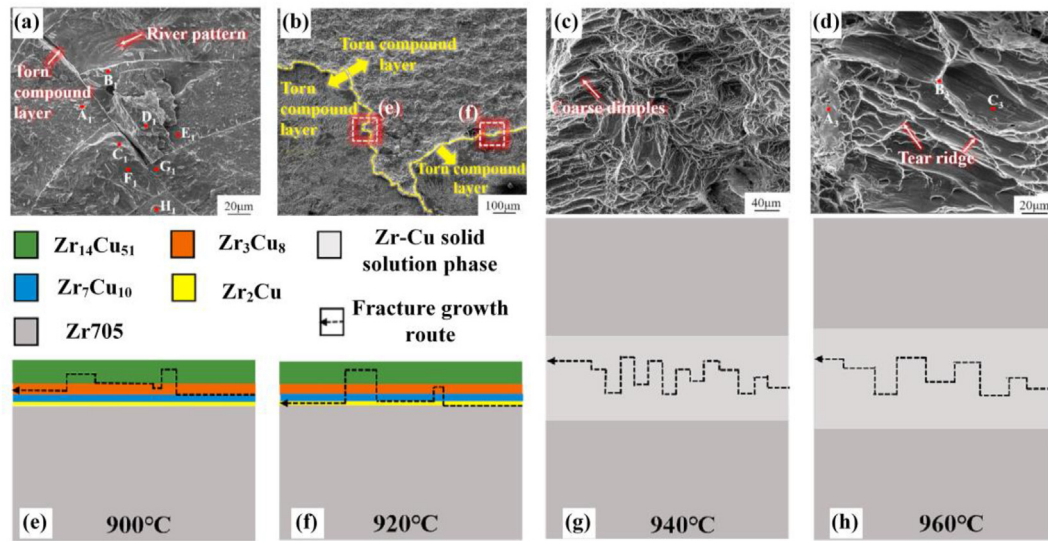


Fig. 9 – SEM micrographs of the fracture morphology and Schematic diagrams of the failure processing of the diffusion-bonded joints at various temperatures: (a) and (e) 900 °C, (b) and (f) 920 °C, (c) and (g) 940 °C, (d) and (h) 960 °C [51].

that the base metals are not melted. In order to ensure the high property of the joint, temperature, holding time, selection of brazing filler, design of joint and other conditions that affect the performance of joint need to be discussed [76,77].

3.1. Influence of brazing filler on the properties of joint

The melting and wetting of the brazing filler is crucial during brazing, so the choice of the filler requires that it must be able to wet the base metal, In addition, the liquidus of filler is

required to be higher than 450 °C but lower than the melting point of the base metal alloy [78]. There are many kinds of brazing fillers that can be used for brazing zirconium alloys. The selection criteria for brazing fillers are generally related to metallurgical compatibility between the base metal alloy and the brazing fillers, operating conditions including working temperature and environment [79].

Lee et al. proposed a new type of filler alloy, Zr(Ti)–Cu(Fe)–Al alloy, for brazing zirconium alloy [80]. Part of Zr in ZrCuAl ternary alloy is replaced by Ti, and part of Cu is replaced by Fe,

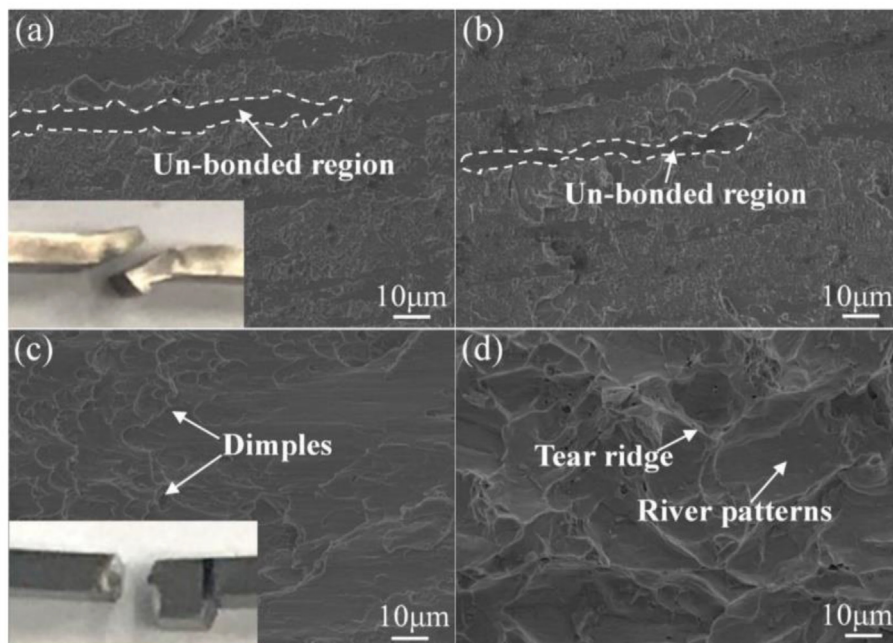


Fig. 10 – Fracture morphology of joints bonded at different temperatures (7 MPa/60 min): (a), (b) 760 °C, (c) 800 °C and (d) 820 °C [66].

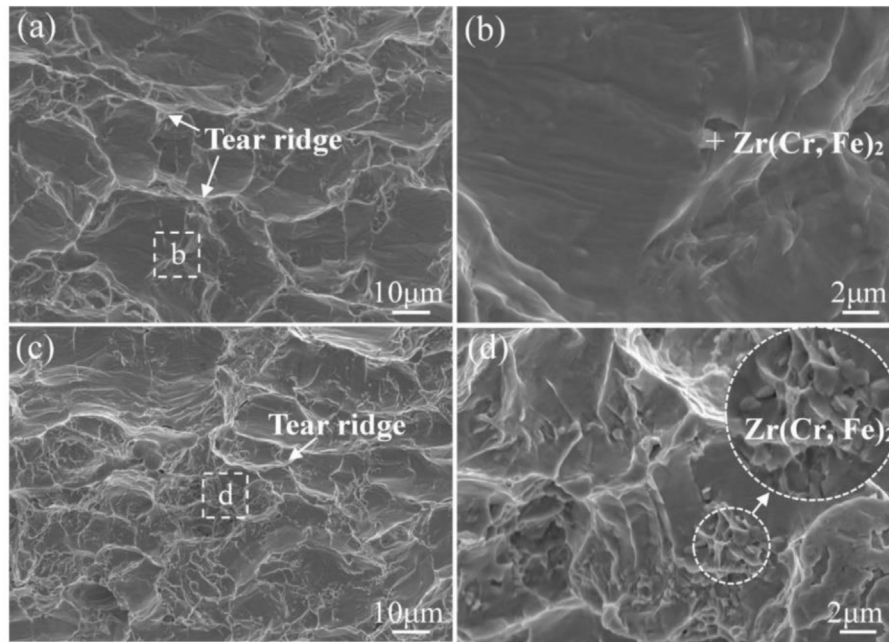


Fig. 11 – Fracture morphology of bonded joints with different holding time (800 °C/7 MPa): (a, b) 30 min and (c, d) 90 min [66].

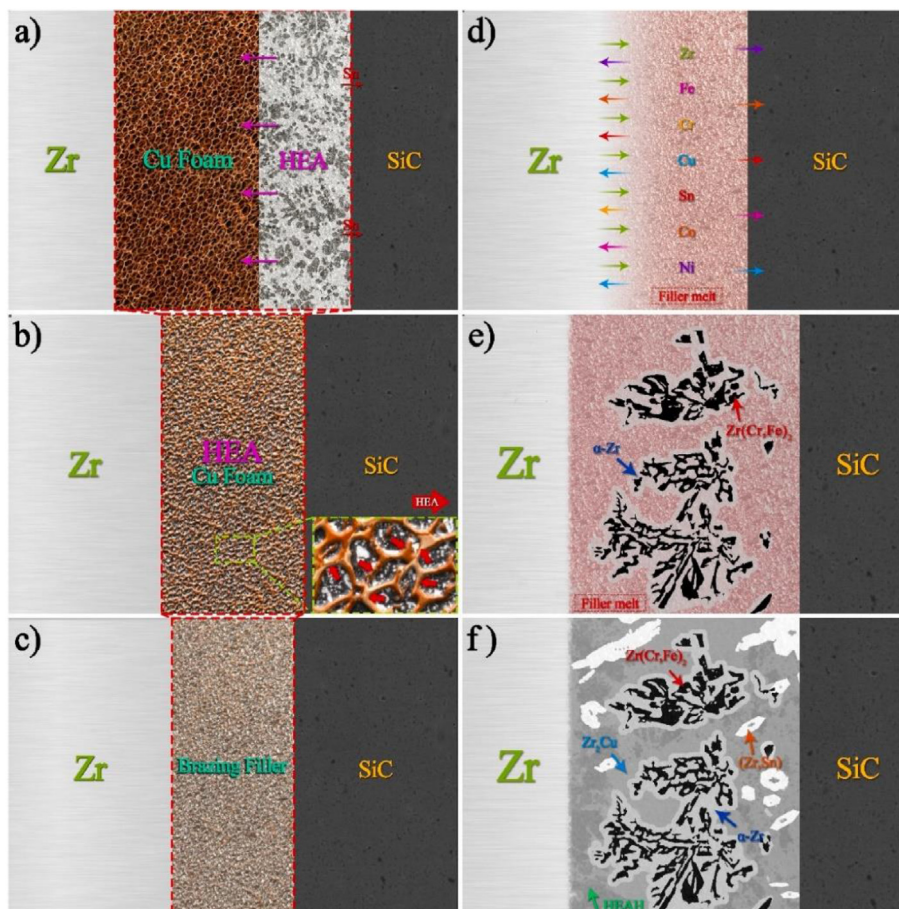


Fig. 12 – Schematic diagram of microstructure evolution of SiC/CoFeCrNiCuSn/Cu/Zr brazed joint [81].

which becomes a five-element alloy. This alloy shows good compatibility with zirconium alloy base material. For brazing Zr and SiC, Luo et al. prepared CoCrFeNiCuSn high entropy alloy and formed a composite interlayer with Cu [81]. Fig. 12 shows the microstructure evolution of brazed joints. The main phases in the joint were identified at different brazing temperatures to be a high-entropy alloy phase, α -Zr (s, s), Zr_2Cu , (Zr, Sn) and $Zr(Cr, Fe)_2$ Laves phase. The joint brazed at 1040 °C for 20 min exhibited a maximum shear strength of 221 MPa at room temperature and an average shear strength of 207 MPa at 600 °C. The room temperature and high temperature-strength obtained here are much higher than those obtained for joints brazed using a conventional filler. Owing to the high-entropy effect, the joint matrix is mainly composed of solid solution phase, which improves the strength and thermal stability of the joint. The existence of the hard $Zr(Fe, Cr)_2$ Laves phase and the soft α -Zr (s, s) phase in the joint significantly improves its strength and plasticity.

Zircaloy-4 was successfully brazed to SiC ceramic using Zr-24Ni (at.%) filler alloy for nuclear application [82]. The maximum shear strength of the joint was 110 MPa at a brazing parameter of 1060 °C for 15 min due to the moderate interface reaction between filler and substrates. When the irradiation fluence was higher than 5×10^{16} ions/cm², the transformation of SiC and Zr_2Ni from crystalline into amorphous state was observed. As the irradiation fluence increased, microcracks appeared in SiC and ZrC due to the uncoordinated irradiation swelling between them.

Arenas et al. [83] found that the slow heating rate of filler metal with wide melting range may lead to the equilibrium between solid phase and liquid phase and coexist of them for a long time. The effect will make the liquid flow into the joint gap and separate from the solid residue, and the chemical inhomogeneity produced in the process will be unfavorable to the strength of the joint.

3.2. Influence of processing parameter on properties of joint

The main process parameters in the brazing process are brazing temperature and holding time, which directly affects the interaction process between the brazing material and the base material. Therefore, it is decisive for the performance of

the joint. The main basis for determining the brazing temperature is the melting point of the brazing filler. The brazing temperature should be appropriately higher than the melting point of the brazing filler to reduce the surface tension of the brazing material so that the brazing material and the base material can fully interact with each other and improve the strength of the joints. Therefore, in general, the brazing temperature is chosen to be 25 °C–60 °C above the liquid phase line temperature of the brazing material [84].

Determining the brazing holding time requires consideration of the characteristics of the brazing material interaction with the base material. When the interaction between the brazing material and the base material has adverse effects such as strong dissolution and intergranular penetration, it is very important to reduce the holding time appropriately. On the contrary, when the interaction between the brazing filler and the base material can eliminate the low-melting organization or brittle phase in weld [84], the holding time should be extended. In addition, the holding time is also related to the size of the weld and the value of the weld gap, when the weld gap is large, in order to ensure the necessary interaction between the brazing material and the base material, it is necessary to extend the holding time.

In order to further understand the influence of brazing temperature on the properties of zirconium alloy joints, many scholars have carried out experiments to explore. Fang et al. [82] studied the influence of brazing temperature on the shear strength of Zircaloy-4/SiC joints. With the increase of brazing temperature from 1030 °C to 1060 °C, the shear strength of the joints increased from 32 MPa to 110 MPa. With the continuous increase of brazing temperature, the shear strength of the joints began to decrease from 110 MPa to 95 MPa. Fig. 13 (a) shows the influence of Zircaloy-4/SiC brazing temperature on the shear strength of the joints. Similarly, Lin et al. [85] controlled several groups of welding temperatures to study their effects on the properties of zirconium alloy joints. The results showed that the shear strength of the joints was at the same level, that is, 240 MPa–300 MPa at the brazing temperature below 1100 °C. When the temperature is higher than 1150 °C, it is found that the shear strength of the joints increases significantly, and the shear strength of the joints reaches the maximum value of 365 MPa at 1200 °C. Tian et al. [86] tested the ZrB_2 -SiC-C/GH99 joints, the final result was

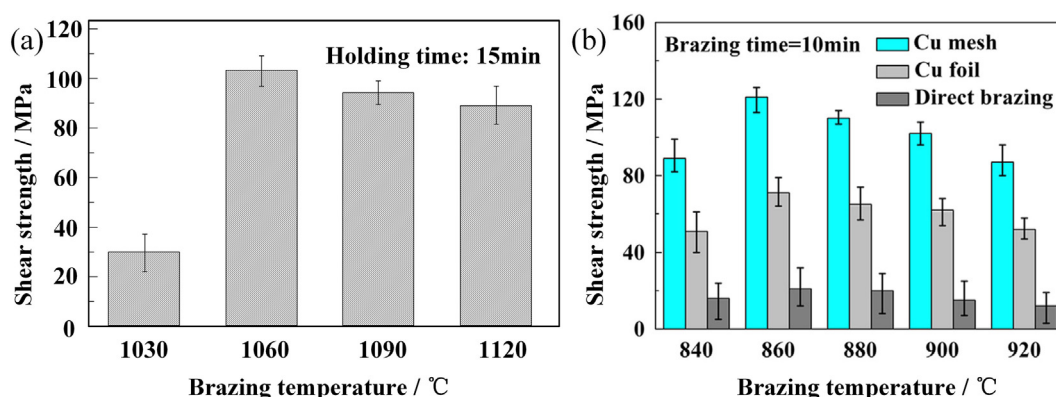


Fig. 13 – Effect of brazing temperature on shear strength of joints: (a) Zircaloy-4/SiC joints [82] (b) ZrB_2 -SiC-C/GH99 joints [86].

similar to that of the above two researchers. With the increase of brazing temperature, the shear strength of the joints first increased and then decreased, and the scholar also studied the influence of the interlayer metal and the shape change of the interlayer metal on the joint strength. The final result was that the shear strength of the joints reached the maximum at 860 °C, and decreased with the increase of brazing temperature, as shown in Fig. 13 (b). In order to fully obtain the changing law of the strength of zirconium alloy joints, Lee et al. [87] conducted experiments on the tensile strength of zirconium alloy joints. They controlled the brazing temperature at 880 °C, 900 °C, 920 °C. The final results showed that the tensile strength of the joints gradually increased, with the tensile strength of 470 MPa at the lower temperature of 880 °C, 513 MPa at the middle temperature of 900 °C, 920 °C at the higher temperature.

Regarding the factors of brazing temperature affecting the joint strength, Fang et al. [82] concluded that the brazing temperature mainly affects the joint strength by influencing the interfacial bonding, residual stress, and the thickness of the reaction layer. Firstly, the interfacial reaction is enhanced when the brazing temperature increases [88,89], which significantly improves the joint strength. Secondly, for the shear fracture mechanism of the joints, the residual stresses should be considered [90]. The difference in thermal expansion coefficients between dissimilar materials leads to residual stresses in the joints [84,89]. When the brazing temperature is higher than 1060 °C, the fracture is mainly located inside the SiC, which indicates that the joint fracture is caused by residual stresses. Finally, for the thickness of the reaction layer, when the temperature increases from 1060 °C to 1120 °C, this may cause an increase in the thermal mismatch of the joints due to the thickening of the reaction layer, resulting in a decrease in the shear strength of the joints.

Regarding the influence of holding time on the properties of zirconium alloy joints, Liu et al. [91] kept the brazing temperature at 910 °C and controlled different holding times. With the increase of holding time, the shear strength of ZrB₂-SiC/Ti6Al4V joints increased, reaching 254 MPa at 600s, reaching a peak of 345 MPa at 1200s, and then decreasing, reaching

262 MPa at 1800s and 210 MPa at 3000s. Fig. 14(a) shows the effect of holding time on the shear strength of ZrB₂-SiC/Ti6Al4V joints. Lin et al. [85] increased the holding time at different brazing temperatures. It was found that the shear strength of Zircaloy-2 joints increased with increasing holding time at different brazing temperatures could improve the shear strength of the joints. Fig. 14(b) shows the influence of holding time on the shear strength of Zircaloy-2 joints. Zhang et al. [92] studied the influence of holding time on the shear strength of ZrB₂-SiC-C/GH99 joints, and the final result was similar to that of Liu et al. [91], that is, at a fixed brazing temperature, with the increase of holding time, the shear strength of the joints first increased and then decreased.

Liu et al. [91] suggested that the low strength of joints at short holding times can be explained by atomic diffusion. Low holding times lead to low interfacial atomic diffusion, resulting in insufficient interfacial bonding between the brazing material and the base material [84]. Long holding times generate large residual stresses at the joint interface due to the mismatch between the Young's modulus and thermal expansion coefficient of the two materials [92]. On the other hand, a longer holding time causes the formation of brittle compounds at the interface, which weakens the plastic deformation of the joints, which is not conducive to the relaxation of the residual stresses in the joints and eventually leads to the reduction of the joint strength. Zhang [92] and Song [93] mentioned that the insulation time affects the thickness of the reaction layer. The thicker reaction layer causes a mismatch between the thermal expansion coefficients of the substrate and the reaction layer, which leads to the weakening of the joint performance.

As for the influence of brazing temperature and holding time on the joint strength, it can be found from the above-mentioned studies that the influence of brazing temperature and holding time are similar. They mainly change the joint strength by affecting the interfacial bonding, residual stress, thickness of the reaction layer, etc. Therefore, in order to obtain high joint strength, the above factors should be fully considered to reasonably control the brazing temperature and holding time, so that the interface elements can fully diffuse and reduce the residual stress at the interface.

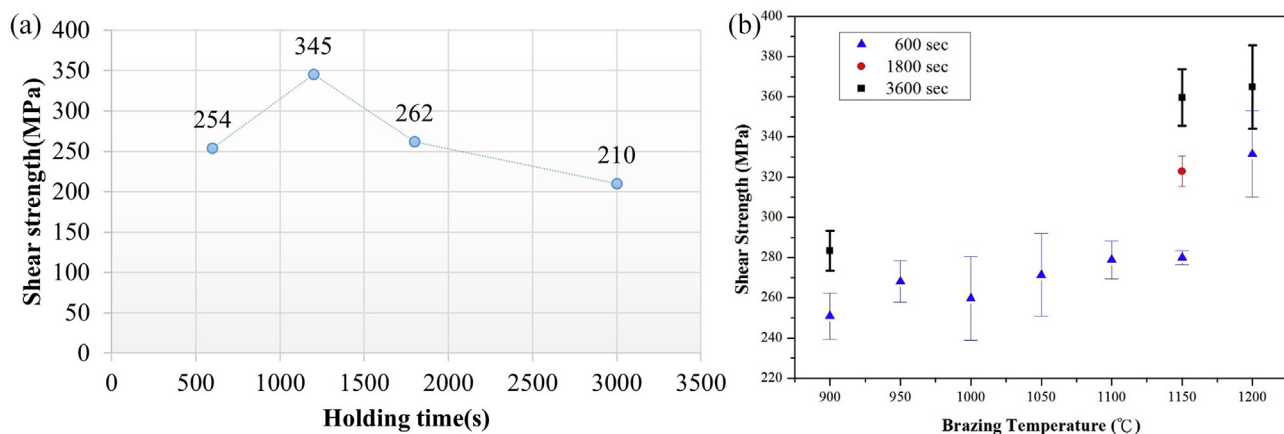


Fig. 14 – Effect of holding time on shear strength of joints: (a) ZrB₂-SiC/Ti6Al4V joints [91] (b) Zircaloy-2 joints [85].

3.3. Influence of other factors on properties of joint

The strength of brazed joints will also be affected by the cleanliness, surface roughness and the gap between the base metals. For the influence of surface pollution, surface pollutants include oil stain, dirt, oxide layer, etc. These surface pollutants will prevent the brazing filler metal from melting and fully welding the base metal. However, although most people think that the cleanliness of the base metal surface has a great influence on the final high-quality joint in the field of brazing, there is little research on this factor. Bobzin team have studied the pre-cleaning and plasma cleaning of the brazing base metal [94]. They found that through these treatments, the surface energy can be increased and the properties of the joint can be improved.

The surface roughness of base metal is an important index to measure the scale deviation of material surface roughness, and it is one of the most common parameters that affect the performance of brazed joints. Most studies have found that smooth surface can provide better shear strength of joints to some extent. Chen et al. studied the influence of surface roughness of base metal on brazing filler [95]. It is found that the wettability of brazing filler decreases with the coarsening of the base metal. Hitchcock and others also found the same conclusion [96]. The roughness of base metal surface leads to the decrease of wettability of brazing filler.

Finally, the selection of brazing filler is not only related to the above-mentioned factors, but also related to the requirements of local laws and regulations. For example, in 2012, some European countries banned the use of cadmium-containing filler in food industry, pharmaceutical industry and dairy industry [97]. Due to the toxicity problem, some elements are not allowed to exist in biomedical materials, such as a high level of Cu^{2+} is considered to be cytotoxic to stem cells [98,99].

4. Resistance welding of zirconium alloys

Resistance welding is a welding method which uses the resistance heat generated by the current passing through the weldment and contact as the heat source to heat the local part and need to be pressurized at the same time. This method has the advantages of no metal filling, high production efficiency, small weldment deformation and easy automation, so it has been applied in many welding fields. Among them, the methods of resistance welding can be mainly divided into four categories, namely spot welding, seam welding, projection welding and butt welding. This review focuses on the resistance butt welding of zirconium alloy. The heating process of resistance butt welding is fast, and the probability of metal reacting with air at the joint is small, so resistance butt welding generally does not need to be completed under shielding gas. Compared with other welding methods, resistance butt welding has the advantages of low heat input, small grain size at the weld, and excellent mechanical properties of the joint. In addition, there are few defects in the joints obtained by resistance butt welding, and there are no welding defects such as air holes and slag inclusion. At present,

resistance butt welding is a widely used welding method for zirconium alloy end plugs and clad tubes.

At present, nuclear power is recognized as a safe and efficient green energy that can replace traditional thermal power. However, the safety of nuclear power is an important factor affecting the healthy development of nuclear power industry. In recent years, a number of nuclear accidents have aroused fears about the reactor core. Generally speaking, nuclear power plants have three physical barriers. The first barrier is the cladding of nuclear fuel rods, which ensures that the fuel assemblies operate well in a sealed environment. The second barrier is the pressure shell, and the third barrier is the containment. Among these three barriers, the inner wall of the nuclear fuel rod cladding is exposed to the pressure of the fuel assembly and fission-generated gas, and the outer wall of the cladding is eroded and corroded by coolant. Therefore, the cladding of the nuclear fuel rod is the most demanding part of the reactor and the most vulnerable to damage. The fuel rod is composed of clad tube, end plug, UO_2 pellet and spring, in which the clad tube and end plug are connected by welding to seal and isolate the nuclear fuel. At present, the main clad tube and end plug materials are zirconium alloy, austenitic stainless steel and silicon carbide composites [100]. With low thermal neutron absorption cross section, excellent corrosion resistance and high temperature mechanical properties, zirconium alloys are the most widely used core structure material including nuclear reactor fuel rod cladding material. The clad tube and the end plug are connected by welding, and the welding joints are the most likely area of potential hazards. The main welding methods of zirconium alloy end plugs and clad tubes are tungsten gas shielded welding, laser welding, electron beam welding and resistance butt welding, but among many welding methods, resistance butt welding has the advantages of stable welding quality and high efficiency. However, due to technical confidentiality and other factors, fewer studies on resistance butt welding of zirconium alloy fuel rod clad tubes and end plugs have been reported by researchers.

4.1. Main parameters of resistance butt welding

The end plug and the clad tube are respectively clamped by the electrodes and assembled into the butt joints. Under the action of cylinder pressure, the end plug is in close contact with the clad tube, and at the same time, the current flows from the end plug side to the tube side through the electrode. The contact resistance at the joint and the body resistance of the workpiece will generate resistance heat, and under the combined effect of force and heat, the joint will quickly reach a plastic state (or molten state) and form a joining.

The main parameters of resistance butt welding of fuel rod cladding materials include welding current, pressure, contact resistance between end plug and electrode and clad tube, welding time, etc., in which welding current, contact resistance between end plug and electrode and clad tube are the main parameters, and welding time is generally 20–50 ms. The geometric feature of resistance welding of cladding material is the plastic deformation of welding interface, which depends on these process parameters and the initial geometry

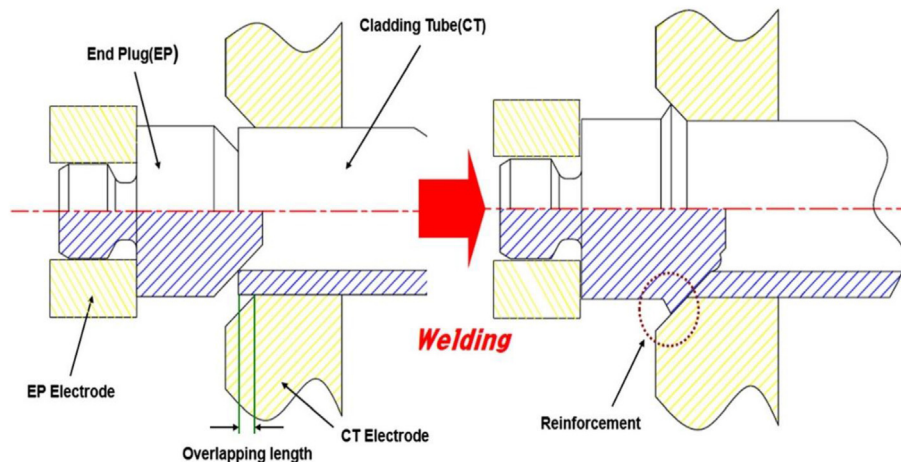


Fig. 15 – Schematic diagram of resistance pressure welding [101].

of the workpiece. The microstructure evolution of the joint and the shape and size of the heat affected zone are also the results of these parameters. The schematic diagram of the resistance butt welding method between the end plug and the clad tube is shown in Fig. 15.

4.1.1. Welding current

Welding current is the most important factor affecting the joint forming during resistance welding [102–104]. If the welding current is too small, the actual temperature at the joint will be too low [105], so that the joint cannot reach the plastic state, and the clad tube and the end plug cannot form an effective welding. Excessive welding current and high welding temperature will easily lead to serious metal oxidation near the joint, increasing the width of heat affected zone and reduce the comprehensive properties of the joint.

During resistance welding, the investigating on the welding current and mechanism can be carried out by simulation and experiments. The simulations were carried out to identify the mechanism of resistance welding of the fuel rod and to analyze the welding phenomena according to the current conditions. Simulation shows the welding time for 1 cycle. The variation of welding current with time during the process of resistance butt welding between cladding tube and end plug is shown in Fig. 16 [106]. The simulation results are divided into five sections, A to E, and are verified in each section. Each section is defined as follows; 1st half cycle: current start point (A)/current peak point (B)/cooling time point (C), 2nd half cycle: current peak point (D)/current fall point (E). The temperature nephogram corresponding to each point is shown in Fig. 17 [106]. In stage A-B, the end plug and the cladding tube form a close contact under pre-pressure, and then a large welding current is rapidly applied to rapidly generate heat in the joint to reach a plastic (or molten) state, and the workpiece forms a preliminary joint under welding pressure in stages B and C; In stages D and E, the joint will further form a connection, and the molten material will gradually cool down and generate new tissues. Although there is still current passing through this stage, the contact resistance at the joint will not increase its temperature. Fig. 18 shows the cross-

section comparison between the actual and simulation weldment. The shape and angle of the fusion zone are almost the same. It can be seen that the melted volume of the weld obtained through the simulation is almost equal to the actual value obtained from the experiment.

An appropriate temperature field and an accurate prediction before the upsetting process for it are desired to ensure the quality of joints. Wang et al. [107] established the two-way coupling model of flash butt welding (FBW) electric heating of AC rail by a combination of finite element simulation and experiment to study the influence of welding current distribution on the quality of joint. It is found that the welding current affects the joint quality by changing the temperature field of resistance butt welding. In order to understand the forming process deeply, an axisymmetric electro-thermal-mechanical process finite element model was established by Feng et al. [108]. Due to the contact resistance, a concentrated heat flow was rapidly generated on the tube-plug contact surface caused by welding current.

Most of the numerical simulation focuses on the distribution and evolution of temperature field and welding current in the welding process. It is suggested researchers pay more attention on simulation work about the effect of welding current on the microstructure and properties of resistant welded joints, which will make accurate prediction and save more costs.

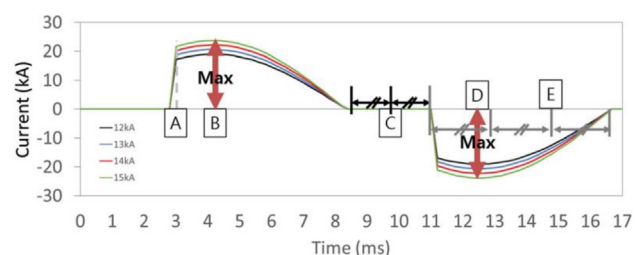


Fig. 16 – Relationship between welding current and time [106].

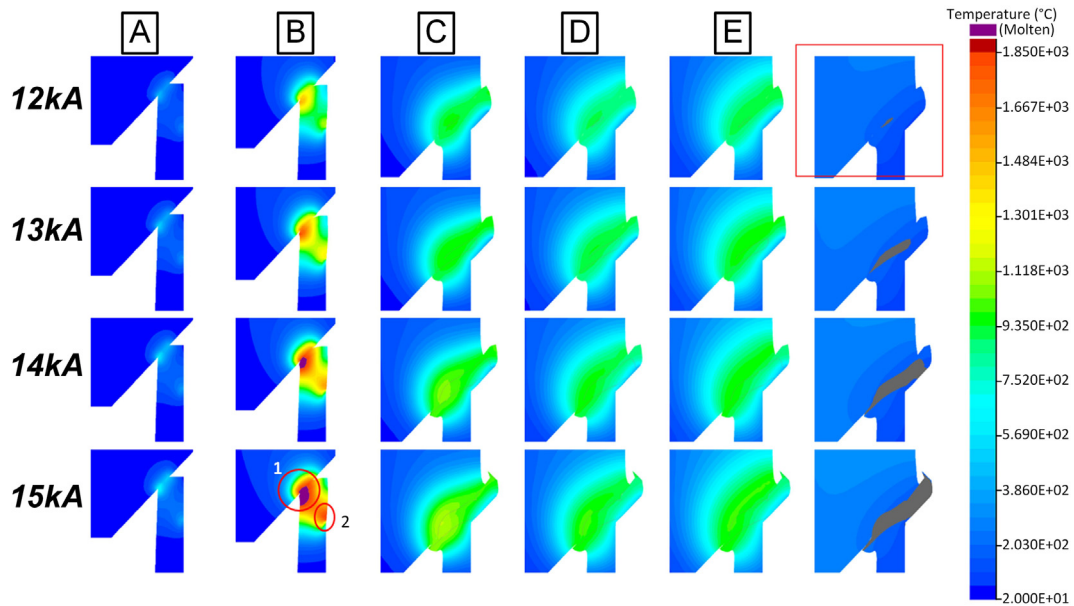


Fig. 17 – Cloud image system of welding temperature distribution [106].

4.1.2. Contact resistance

The contact resistance between the end plug and electrode, the clad tube and electrode, and the end plug and the clad tube are the main factors to produce welding heat source in resistance welding. Among them, the contact resistance between the end plug and the clad tube has an important influence on the forming process of the joint, but during the welding process, the contact resistance varies with pressure, temperature and the completion of the connection process, which is difficult to be measured directly by experimental methods. Doyen studied a method of calibrating contact resistance by using finite element model [109]. The feasibility of this method was verified by experiments. It was also found that in the resistance welding of end plug and pipe, the deformation caused by improper parameters all appears on one side of the pipe.

The length of the protruding electrode of the clad tube affects the contact resistance between the end plug and the electrode, and between the cladding tube and the electrode, thus affecting the formation of the joint, Doyen [110] and Corpac [111] performed resistance welding tests on the end plug and cladding tube made of oxide dispersion strengthened steel. It is pointed out that a high deformation zone will appear near the contact between the clad tube and the electrode. During welding with high heat input, it will cause the pipe to collapse. The root cause of the collapse is that there is a large current density between the clad tube and the electrode. The researchers found that this defect could be avoided by changing the extension length of the clad tube. The specific value of the contact resistance and its variation with parameters during resistance butt welding of fuel rod cladding material needs to be further investigated.

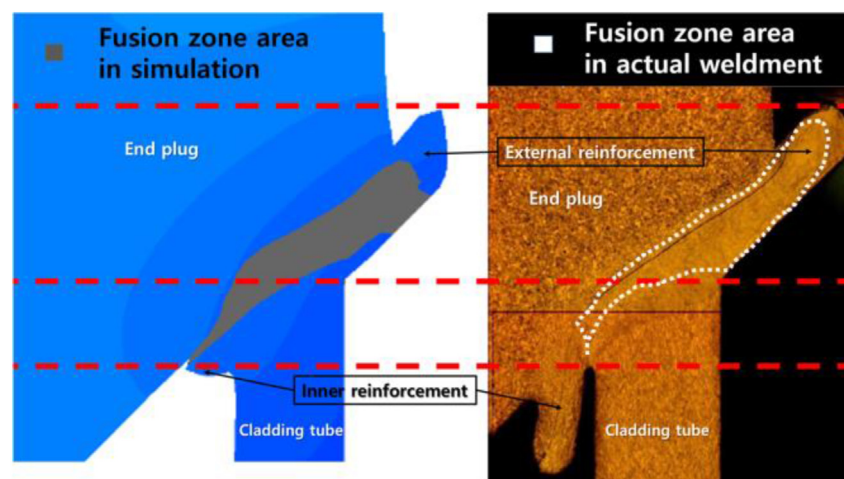


Fig. 18 – Shape comparison between simulation and experiment cross section [106].

The contact resistance between the end plug and the clad tube has an important influence on the formation of the joint. Through the analysis of the contact resistance, the connection mechanism of resistance welding of cladding materials can be understood more deeply. However, it is difficult to obtain the value of contact resistance by experimental method, and the variation trend of contact resistance with process parameters in the welding process is not clear.

In addition, a few scholars have studied the influence of welding pressure on resistance and welding head. Kersten obtained the heat distribution during resistance upsetting butt welding through finite element simulation [112] and the results were verified by experiments. The results show that due to the welding pressure, the heat distribution at the joint is uneven, the temperature inside the weld is lower, while the temperature on both sides is higher.

From the above, it can be seen that the key problem of connecting zirconium alloy end plugs and clad tubes by resistance welding is the reasonable matching of several important welding parameters. The above scholars have made clear the technological process, welding parameters and defect causes of resistance welding of end plugs and clad tubes through experiments, but there is a lack of in-depth discussion on the welding mechanism and the influence of welding parameters on the change of microstructure.

Most of the scientific research is about the technological parameters of resistance butt welding of zirconium alloy, but there is little research on the microstructure of welded joints. The relationships among phase composition, grain size and morphology, grain orientation and mechanical properties of the joints need to be further studied. The influence of welding parameters on the corrosion resistance of joints is also unclear.

4.2. Microstructure and mechanical properties of the joint

During resistance welding of zirconium alloy joints, cooling rate has significant effect on the microstructure of joints. Martensite phase is formed at the weld when cooling at a faster speed, and there is no obvious alloy element segregation. The process conditions of rapid heating and rapid cooling will also reduce the amount of dissolved nitrogen in the weld metal, so that the weldment has good corrosion resistance. Under the condition of slow cooling, the widmanstatten microstructure will be formed in the weld, which will adversely affect the properties of the joint [111,113,114]. The cooling rate of resistance butt welding of Zr-4 alloy is about 103 °C/s, which is a high cooling rate. The martensite microstructure and fine widmanstatten microstructure were found in the weld zone and heat affected zone [115]. Kumar et al. [113] studied Microstructural studies and crystallographic orientation of different zones and δ -hydrides in resistance welded zircaloy-4 sheets. As shown in Fig. 19 (a), in the central weld zone, colonies of α -phase with no residual β -phase are present. The weld microstructure is characterized by the presence of fine acicular α -grains inside the colonies in the form of basket-weave microstructure. These colonies are within the prior- β -grains. The microstructure in the heat affected zone is characterized by the presence of recrystallized α -grains with grain size ranging from sub-micron to about 3 μm . Crystallography of δ -ZrH_{1.5} hydride in the weld and heat affected zone was shown in Fig. 20. The hydride platelets were mainly observed inside prior β -grains, where the lamella of the α grains have the angular differences of 10°, 60–63° and 90°. These hydride platelets are present both inside the grains and at grain boundaries of the prior β -grains.

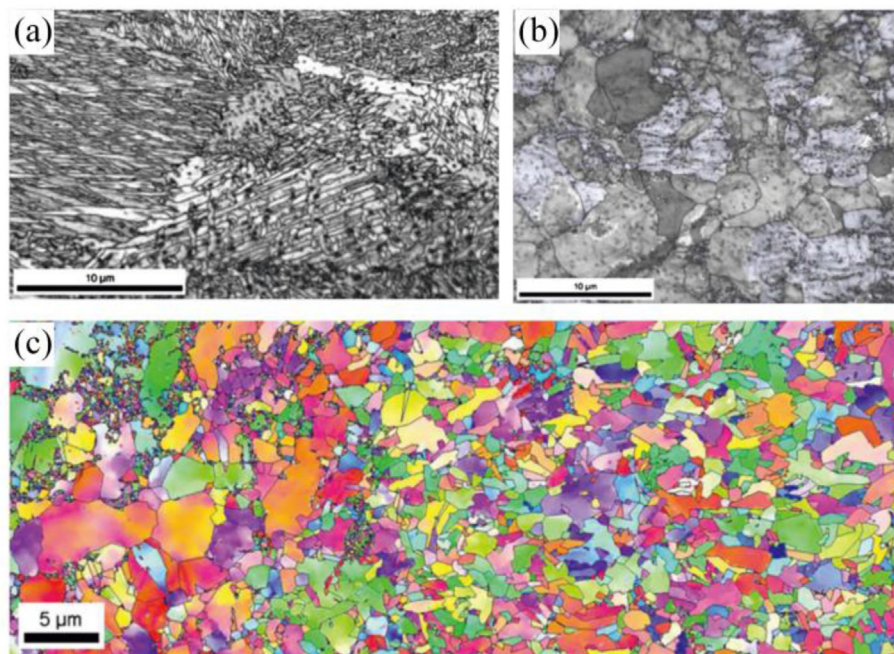


Fig. 19 – Microstructure of central weld zone and heat affected zone: (a) the central weld zone. (b) heat affected zone (c) as-received zone [113].

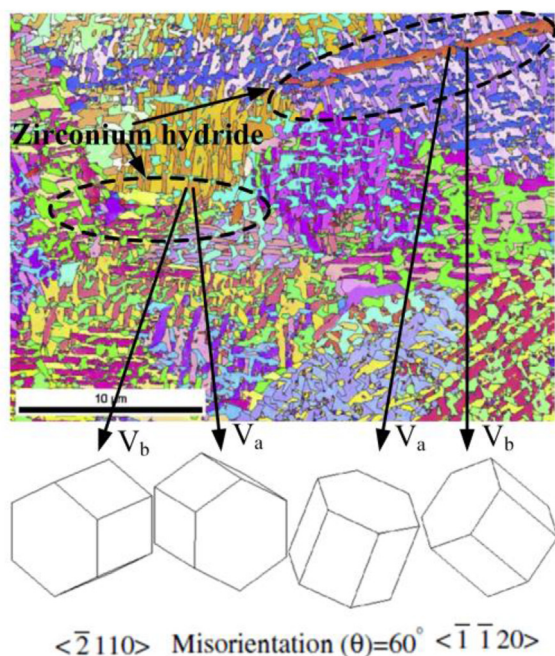


Fig. 20 – Crystallographic orientation of hydrides analyzed using $\{0\ 0\ 1\}$ $\{1\ 0\ 1\}$ and $\{1\ 1\ 1\}$ pole figures of the matrix and the hydride grains [113].

Most of the hydride platelets grow perpendicular to alternating α -lamellas having an angular difference of 60–63° as shown in Fig. 20.

At present, there are few reports on the analysis of microstructure characteristics of zirconium alloy resistance butt welding and the influence of welding parameters on microstructure, such as the influence of parameters on grain size, poor grain boundary orientation and phase distribution. The mechanism of the influence of the microstructure of resistance butt welding joints of zirconium alloys on the corrosion resistance and mechanical properties of the joint remains to be further investigated.

Under the metallographic microscope, it can be observed that the joint obtained by resistance butt welding has a very fine dividing line, which is the fusion line between the end plug and the clad tube. One of the evaluation criteria of a joint with excellent performance is that the length of the fusion line is greater than the wall thickness of the clad tube. At present, the formation mechanism of fusion line is not clear.

Park et al. [116] found that there was a black substance in the grain in the metallographic structure after welding and the test showed that the black substance was zirconium carbide phase. The precipitated zirconium carbide had little influence on the strength of the joint, but had some influence on stress corrosion. The researchers detected the quality of the weld by monitoring instantaneous dynamic resistance, instantaneous welding pressure and electrode displacement, and found the factors suitable for monitoring the quality of the weld through a series of simulation tests [117].

There is no obvious fusion zone in the resistance butt welded joint, only a very thin fusion line is formed, Na et al. [106] carried out numerical simulation and blasting test on the

zirconium alloy cladding material connector obtained by resistance welding. The simulation results show that the length of the molten structure at the joint was greater than the thickness of the tube wall. The influence of process parameters on the formation and strength of the joint was also discussed. The blasting test results showed that the fracture occurred at the base material of the clad tube, not at the weld. The micro-hardness of the joint is higher than that of the base materials. The higher welding current and extended length will improve the strength. The welding pressure has little influence on the test results, and the welding current and extended length have significant influence on the weld quality. French combined the finite element model of resistance butt welding process with the blasting test model to simulate the high-quality weld and poor-quality weld in the test [118], so as to optimize the welding quality and design a new welding structures.

The crack strength and hardness of resistance butt welded joint are higher than that of the base metal. The strengthening mechanism of zirconium alloy base metal is mainly fine grain strengthening and second phase particle strengthening. However, the results of microstructure analysis of the joint show that only a small amount of second phase particles exist in the weld and heat affected zone. At present, the state of existence of second-phase particles after welding and the strengthening mechanism of the welded joint still need to be further studied. At the same time, the performance test method of the welded joint between the composite tube and the end plug is relatively simple. In addition to the nondestructive testing methods, the appropriate nondestructive testing method can be applied to the performance testing of the joint, and the corresponding evaluation standards can be formulated.

5. Conclusion

Based on the summary and review on the diffusion bonding, brazing and resistance welding of zirconium and its alloys, the following conclusions can be drawn.

Firstly, for the diffusion bonding of zirconium alloys, the process parameters should be controlled and optimized to obtain reliable joints. The selection of the interlayer is significant for controlling the thickness of interfacial reaction products to improve the bonding strength, which includes the type, thickness, composition of interlayer. In some cases, for economic cost and other factors, it is possible to obtain a good joint of zirconium alloy without interlayer. Therefore, the engineering requirements for diffusion bonding zirconium alloys should be considered individually when selecting the interlayer. Interfacial microstructure and diffusion behavior has important influence on the properties and defects of joints. In a word, the process-microstructure-property relationship during diffusion bonding zirconium alloys is imperative and is also a crucial tool for selecting appropriate process parameters.

Secondly, for brazing zirconium alloys, the selection of brazing filler and the optimization of the brazing temperature and holding time has important effect on the quality of brazed joints. When selecting brazing filler, the chemical composition, melting point and fluidity need to be considered. In determining the brazing temperature and holding time, factors such

as the coefficient of thermal expansion and thermal conductivity of zirconium alloy need to be fully considered, and also need to be optimized according to the specific brazing requirements and the characteristics of the brazing filler. The brazing conditions should be optimized according to the recommended brazing temperature and holding time on a case-by-case basis. Future research directions include further optimization of the brazing process parameters, exploring the relationship between the microstructure and property of brazed joints, and developing new brazing filler to improve the performance and application of brazing zirconium alloy. With the development of new technology, integrated computational materials engineering and machine learning in material design can be used for designing and optimizing the brazing process of zirconium alloys.

Finally, the quality of zirconium alloy resistance welded joints can be significantly improved by optimizing the welding current and contact resistance by avoiding overheating and overmelting. In practical applications, it is recommended to use a combination of process methods to realize efficient resistance welding of zirconium alloys. Numerical simulation method can be used to predict the welding results and simulate the physical mechanism of resistance welding process and welding defects. It is suggested to do more work on developing special resistance welding equipment for specific application in the field of nuclear. The stability and optimization of processing parameters of the resistance welding is also needed more attention in the future research. In addition, the standardization and specification about resistance welding zirconium alloy needs to be strengthened to ensure the quality and stability of welded joints.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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